MATHEMATICS — CLASS X

TRIGONOMETRY

Trigonometric Ratios | Standard Angles | Complementary Angles | Identities
Unit Test | Session 2025–26 | CBSE Syllabus (Ch. 8)

1. Use of calculators is NOT permitted. Standard values: $\sin 30° = 1/2$, $\cos 60° = 1/2$, $\tan 45° = 1$, $\sqrt{2} \approx 1.414$, $\sqrt{3} \approx 1.732$.

SECTION A — Multiple Choice Questions (1 × 10 = 10 Marks)

Choose the correct answer. Each question carries 1 mark.

Q1. If $\sin A = 3/5$, the value of $\cos A$ is: [[1]]
(a) ¾ **(b)** ⅘ **(c)** 5/3 **(d)** 4/3

Q2. The value of $(\sin^2 60° + \cos^2 60°)$ is: [[1]]
(a) 0 **(b)** ½ **(c)** $\sqrt{3}/2$ **(d)** 1

Q3. The value of $\tan 45° + \cot 45°$ is: [[1]]
(a) 1 **(b)** $\sqrt{2}$ **(c)** 2 **(d)** 0

Q4. If $\cos A = 12/13$, then $\tan A =$ [[1]]
(a) 5/12 **(b)** 12/5 **(c)** 5/13 **(d)** 13/5

Q5. The value of $(1 - \sin^2\theta)$ is equal to: [[1]]
(a) $\sin^2\theta$ **(b)** $\tan^2\theta$ **(c)** $\cos^2\theta$ **(d)** $\cot^2\theta$

Q6. $\sin(90° - A)$ is equal to: [[1]]
(a) $\sin A$ **(b)** $-\sin A$ **(c)** $\cos A$ **(d)** $-\cos A$

Q7. $\sec^2\theta - \tan^2\theta =$ [[1]]
(a) 0 **(b)** 2 **(c)** −1 **(d)** 1

Q8. The value of $(\sin 30° \times \cos 60°) + (\cos 30° \times \sin 60°)$ is: [[1]]
(a) 0 **(b)** 1 **(c)** ½ **(d)** $\sqrt{3}/2$

Q9. $\text{cosec}^2 A - \cot^2 A$ is equal to: [[1]]
(a) 0 **(b)** −1 **(c)** 2 **(d)** 1

Q10. If $\tan\theta = 1/\sqrt{3}$, then $\cos\theta$ is equal to: [[1]]
(a) ½ **(b)** $\sqrt{3}/2$ **(c)** $1/\sqrt{2}$ **(d)** $\sqrt{3}$

SECTION B — Very Short Answer Questions (2 × 6 = 12 Marks)

Each question carries 2 marks. Show necessary working.

Q11. In a right triangle ABC, right-angled at B, if $\tan A = 1/\sqrt{3}$, find the value of $\sin A \cdot \cos C + \cos A \cdot \sin C$. [[2]]

Q12. Evaluate: $(\sin^2 63° + \sin^2 27°) / (\cos^2 17° + \cos^2 73°)$. [[2]]

Q13. If $\sin\theta + \cos\theta = \sqrt{2}$, find the value of $\sin\theta \cdot \cos\theta$. [[2]]

Q14. Prove that $(1 - \cos^2 A) \cdot \text{cosec}^2 A = 1$. [[2]]

Q15. Find the value of: $(\tan^2 45° - \cos^2 60°) / (\sin^2 30° + \cos^2 30°)$. [[2]]

Q16. If $\sin A = \cos A$, find the value of $2\tan^2 A + \sin^2 A - 1$. **[[2]]**

SECTION C — Short Answer Questions – I (3 × 5 = 15 Marks)

Each question carries 3 marks. Internal choices are indicated.

Q17. Prove the identity: $\sin\theta / (1 - \cos\theta) + \sin\theta / (1 + \cos\theta) = 2\,\mathrm{cosec}\,\theta$. **[[3]] OR** Prove: $(\tan\theta + \sec\theta - 1) / (\tan\theta - \sec\theta + 1) = (1 + \sin\theta) / \cos\theta$

Q18. If $\sec\theta + \tan\theta = p$, prove that $\sin\theta = (p^2 - 1) / (p^2 + 1)$. **[[3]]**

Q19. Evaluate without using trigonometric tables: **[[3]]** $(\sin^2 25° + \sin^2 65°) / (\cos^2 24° + \cos^2 66°) + \sin^2 37° \cdot \mathrm{cosec}^2 53°$ **OR** Prove: $(\mathrm{cosec}A - \sin A)(\sec A - \cos A) = 1 / (\tan A + \cot A)$

Q20. Prove that: $\sqrt{[(\sec A - 1)/(\sec A + 1)]} + \sqrt{[(\sec A + 1)/(\sec A - 1)]} = 2\,\mathrm{cosec}A$. **[[3]] OR**

If $\tan A = n \cdot \tan B$ and $\sin A = m \cdot \sin B$, prove that $\cos^2 A = (m^2 - 1) / (n^2 - 1)$.

Q21. If $\sin\theta - \cos\theta = 0$, find the value of $\sin^4\theta + \cos^4\theta$. **[[3]] OR** Evaluate: $(\sin 35° / \cos 55°) + (\cos 55° / \sin 35°) - 2\cos 60° \div \sin^2 30°$.

SECTION D — Short Answer Questions – II (4 × 4 = 16 Marks)

Each question carries 4 marks. Internal choices are provided.

Q22. Prove the trigonometric identity: **[[4]]** $(\sin A + \mathrm{cosec}A)^2 + (\cos A + \sec A)^2 = 7 + \tan^2 A + \cot^2 A$ **OR** Prove: $(\sin A - 2\sin^3 A) / (2\cos^3 A - \cos A) = \tan A$

Q23. Prove the identity: $(1 + \tan^2 A) / (1 + \cot^2 A) = \tan^2 A$. **[[4]] OR** If $\sin\theta + \sin^2\theta = 1$, prove that $\cos^2\theta + \cos^4\theta = 1$.

Q24. Given that $\tan(A + B) = \sqrt{3}$ and $\tan(A - B) = 1/\sqrt{3}$, where $0° < A + B \le 90°$ and $A > B$, find A and B. Hence evaluate: $\sin(A + B) \cdot \cos(A - B) + \cos(A + B) \cdot \sin(A - B)$. **[[4]]**

OR If $\cos A + \sin A = \sqrt{2}\cos A$, prove that $\cos A - \sin A = \sqrt{2}\sin A$. Hence find the value of tan A.

Q25. Prove the identity: $(\sin A + \cos A)(\tan A + \cot A) = \sec A + \mathrm{cosec}A$. **[[4]] OR** Prove: $(1 + \cot A - \mathrm{cosec}A)(1 + \tan A + \sec A) = 2$.

SECTION E — Long Answer Questions (5 × 3 = 15 Marks)

Each question carries 5 marks. Show all steps clearly. Internal choices are provided.

Q26. Prove the following identities: **[[5]]**

(a) $(\cos A - \sin A + 1) / (\cos A + \sin A - 1) = \mathrm{cosec}A + \cot A$, using the identity $\mathrm{cosec}^2 A = 1 + \cot^2 A$. **(b)** $\sin\theta / (\cot\theta + \mathrm{cosec}\theta) - \sin\theta / (\cot\theta - \mathrm{cosec}\theta) = 2$.

OR

If $\cos\theta + \sin\theta = \sqrt{2}\cos\theta$, show that $\cos\theta - \sin\theta = \sqrt{2}\sin\theta$. Also prove: $\sin^6\theta + \cos^6\theta = 1 - 3\sin^2\theta\cos^2\theta$.

Q27. Prove the following identities: **[[5]]**

(a) $\tan A / (1 - \cot A) + \cot A / (1 - \tan A) = 1 + \sec A \cdot \mathrm{cosec}A$. **(b)** $(\sin\theta + \cos\theta + 1)(\sin\theta - 1 + \cos\theta) \cdot \sec\theta \cdot \mathrm{cosec}\theta = 2$.